

mdutil Manual QA Checklist

Version target: 3.0.0 **Last updated:** 2026-07-27 **Test platforms:** Linux, macOS, Windows (Python 3.11+)

This checklist covers the terminal display viewer, editor, and the v3.0 PDF/HTML export features. Each step includes a concrete command to run and an expected result to verify.

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1. Prerequisites

1.1 Environment

Step	Action	Expected
1.1a	<code>python --version</code>	Python 3.11.x or higher
1.1b	<code>pip install -e .</code> from project root	Installs successfully, no errors
1.1c	<code>python -m mdutil --version</code>	Prints mdutil 3.0.0 (or target version)
1.1d	<code>python -m pytest -q</code>	All tests pass (192 passed at time of writing)

1.2 Test fixture document

Create a file at `/tmp/qa-test.md` with the following content:

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore

2. Basic Viewing & Rendering

2.1 File input

Step	Command	Expected
2.1a	<code>python -m mdutil /tmp/qa-test.md</code>	Opens interactive viewer. All headings, lists, tables, code blocks, blockquotes, and horizontal rules are visible with proper formatting.

2.2 Stdin input

Step	Command	Expected
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| 2.2a | `cat /tmp/qa-test.md \` | `python -m mdutil` | Rendered output prints to stdout and exits. No interactive viewer. | | 2.2b | `cat /tmp/qa-test.md \` | `python -m mdutil -` | Same as 2.2a. |

2.3 Quiet mode

Step	Command	Expected
2.3a	<code>python -m mdutil --quiet /tmp/qa-test.md</code>	No visible output (exit code 0).
2.3b	<code>echo \$?</code>	0

3. Interactive Viewer & Editor

Requires a real TTY. On Windows, use PowerShell or Windows Terminal.

Step	Action	Expected
3.1	<code>python -m mdutil /tmp/qa-test.md</code>	Viewer opens with rendered Markdown. Status bar shows normal mode indicators.
3.2	Press j and k	Scrolls up and down one line at a time.
3.3	Press g then G	g jumps to top, G jumps to bottom.
3.4	Press q	Exits viewer (no changes to save).

3.5	Press i	Enters insert/edit mode. Raw Markdown is shown. Status bar changes to insert indicators.
3.6	Type some text, then Escape	Returns to normal mode. Status bar reverts to normal mode indicators.
3.7	Press dd on a non-empty line	The line is copied to clipboard (no visual change unless pasted).
3.8	Press yy	Copies the current line (same as dd without deleting).
3.9	Press i, move cursor, press Ctrl-V	Pastes clipboard content at cursor.
3.10	Press Escape, then /	Search bar appears at bottom.
3.11	Type Alice, press Enter	First match is highlighted in the rendered preview. Press n / N to cycle forward/backward.
3.12	Press Escape to dismiss search	Search highlight disappears.
3.13	Press q	Returns to normal mode if in search.
3.14	Press Ctrl-S in insert mode	Saves the file. Status bar confirms save.
3.15	Make an edit, press q	Warning: "Unsaved changes. Use !q to discard."
3.16	Press !q	Discards changes and exits.
3.17	Press F1	Help modal opens with keybinding reference. Press any key to close.

4. Themes & Configuration

4.1 Built-in themes

Step	Command	Expected
4.1a	<code>python -m mdutil --theme dracula /tmp/qa-test.md</code>	Opens with Dracula theme colors. Dark background, pink/green highlights.

4.1b	<code>python -m mdutil --theme one-dark /tmp/qa-test.md</code>	Opens with One Dark theme.
4.1c	<code>python -m mdutil --theme high-contrast /tmp/qa-test.md --quiet</code>	Renders to stdout with high-contrast colors.

4.2 Syntax theme

Step	Command	Expected
4.2a	<code>python -m mdutil --syntax-theme monokai /tmp/qa-test.md --quiet</code>	Code blocks use Monokai syntax colors.
4.2b	<code>python -m mdutil --syntax-theme native /tmp/qa-test.md --quiet</code>	Code blocks use native syntax colors.

4.3 Configuration file

Step	Command	Expected
4.3a	<code>python -m mdutil --generate-config</code>	Prints "Configuration file ready: /home/user/.mdutilcfg" (path varies by OS).
4.3b	<code>cat ~/.mdutilcfg</code> (Linux/macOS) or type %USERPROFILE%\mdutil.ini (Windows)	Shows commented INI file with all sections including [export].
4.3c	Edit ~/.mdutilcfg, change theme = dracula, then <code>python -m mdutil /tmp/qa-test.md</code>	Opens with Dracula theme (config default, no --theme flag needed).
4.3d	Restore default theme after test.	

4.4 Line numbers

Step	Command	Expected
4.4a	<code>python -m mdutil --line-numbers /tmp/qa-test.md --quiet</code>	Line numbers shown at the start of each line.

5. PDF Export

5.1 Basic PDF generation

Step	Command	Expected
5.1a	<code>python -m mdutil --export pdf --output /tmp/test.pdf /tmp/qa-test.md</code>	Prints "Exported to: /tmp/test.pdf" to stderr.
5.1b	<code>head -c 5 /tmp/test.pdf</code>	Output starts with %PDF-.
5.1c	<code>file /tmp/test.pdf</code>	Returns "PDF document" (or similar).
5.1d	Open /tmp/test.pdf in a PDF reader (Evince, Chrome, Acrobat).	All content rendered: scaled headings, paragraphs with bold/emphasis/inline-code formatting, clickable links, lists, code blocks (with background fill), tables (bordered, alternating rows, wrapped cell text), blockquote blocks with a vertical rule, horizontal rules.
5.1e	Export mdutil-specification.md and inspect the first page.	The document metadata header entries (Author, Version, Last-Updated, License, Repository) are separate lines, not one collapsed line.

5.2 PDF bookmarks

Step	Command	Expected
5.2a	Open /tmp/test.pdf in a PDF reader with a sidebar/navigation pane.	Bookmarks/outline entries exist for h1, h2, h3 headings. h4-h6 are NOT bookmarked.
5.2b	Click a bookmark.	Jumps to the correct page.

5.3 PDF paper sizes

Step	Command	Expected
5.3a	<code>python -m mdutil --export pdf --output /tmp/test-letter.pdf /tmp/qa-test.md (with pdf_paper_size = Letter in config if testing letter)</code>	PDF generated successfully. File size differs from A4.

5.3b	<code>python -m mdutil --export pdf --output /tmp/test-legal.pdf /tmp/qa-test.md</code> (with config change if testing legal)	PDF generated successfully.
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5.4 PDF with header/footer

Step	Command	Expected
5.4a	Edit <code>~/.mdutilcfg</code> [export] section to add <code>pdf_header = Draft</code> and <code>pdf_footer = Page</code>	
5.4b	<code>python -m mdutil --export pdf --output /tmp/test-hf.pdf /tmp/qa-test.md</code>	PDF generated. Header text "Draft" visible at top of pages, footer text visible at bottom.
5.4c	Restore config to remove header/footer after test.	

5.5 PDF auto-naming (no --output)

Step	Command	Expected
5.5a	<code>cd /tmp && python -m mdutil --export pdf /tmp/qa-test.md</code>	"Exported to: /tmp/qa-test.pdf"
5.5b	<code>ls -la /tmp/qa-test.pdf</code>	File exists.
5.5c	Clean up: <code>rm /tmp/qa-test.pdf</code>	

5.6 PDF via --output-dir

Step	Command	Expected
5.6a	<code>python -m mdutil --export pdf --output-dir /tmp/pdf-out /tmp/qa-test.md</code>	"Exported to: /tmp/pdf-out/qa-test.pdf"
5.6b	<code>ls -la /tmp/pdf-out/</code>	Directory created with PDF inside.

6. HTML Export

6.1 Basic HTML generation

Step	Command	Expected
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6.1a	<code>python -m mdutil --export html --output /tmp/test.html /tmp/qa-test.md</code>	Prints "Exported to: /tmp/test.html" to stderr.
6.1b	<code>head -c 20 /tmp/test.html</code>	Starts with .
6.1c	Open /tmp/test.html in a web browser.	All content rendered: headings, lists, code blocks with syntax highlighting, tables (bordered), blockquotes, links are clickable.

6.2 HTML custom CSS

Step	Command	Expected
6.2a	Create /tmp/custom.css: <code>body { background: #ffe; } h1 { color: red; }</code>	
6.2b	<code>python -m mdutil --export html --custom-css /tmp/custom.css --output /tmp/test-custom.html /tmp/qa-test.md</code>	HTML generated.
6.2c	<code>grep "Custom CSS" /tmp/test-custom.html</code>	Shows /* Custom CSS */ comment in the style block.
6.2d	Open /tmp/test-custom.html in a browser.	Page has cream background (#ffe), h1 headings are red.

6.3 HTML auto-naming (no --output)

Step	Command	Expected
6.3a	<code>cd /tmp && python -m mdutil --export html /tmp/qa-test.md</code>	"Exported to: /tmp/qa-test.html"
6.3b	<code>ls -la /tmp/qa-test.html</code>	File exists.
6.3c	Clean up: <code>rm /tmp/qa-test.html</code>	

7. Multi-Format Export

7.1 Both formats in one pass

Step	Command	Expected
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7.1a	<code>python -m mdutil --export pdf,html --output-dir /tmp/multi /tmp/qa-test.md</code>	"Exported to: /tmp/multi/qa-test.pdf" and "Exported to: /tmp/multi/qa-test.html"
7.1b	<code>ls -la /tmp/multi/</code>	Both files present.
7.1c	Verify qa-test.pdf starts with %PDF and qa-test.html starts with .	

7.2 Stdin with both formats

Step	Command	Expected
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| 7.2a | `cat /tmp/qa-test.md \| python -m mdutil --export pdf,html --output-dir /tmp/multi-stdin` | Exports to /tmp/multi-stdin/output.pdf and /tmp/multi-stdin/output.html. Both valid. || 7.2b | Clean up: `rm -rf /tmp/multi /tmp/multi-stdin` | |

8. Export Edge Cases & Errors

8.1 Unsupported format

Step	Command	Expected
8.1a	<code>python -m mdutil --export docx /tmp/qa-test.md</code>	"Error: Unsupported export format: 'docx' (choose from pdf, html)" — exit code 1.

8.2 Missing input file

Step	Command	Expected
8.2a	<code>python -m mdutil --export pdf nonexistent.md</code>	Error about file not found. Exit code 1.

8.3 Read-only output directory

Step	Command	Expected
8.3a	<code>mkdir -p /tmp/readonly && chmod 444 /tmp/readonly</code>	
8.3b	<code>python -m mdutil --export pdf --output-dir /tmp/readonly /tmp/qa-test.md</code>	"Export error (pdf): Permission denied — cannot write to /tmp/readonly/qa-test.pdf"

8.3c	<code>chmod 755 /tmp/readonly && rmdir /tmp/readonly</code>	Clean up.
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8.4 Missing custom CSS file

Step	Command	Expected
8.4a	<code>python -m mdutil --export html --custom-css /tmp/nonexistent.css /tmp/qa-test.md</code>	"Error reading custom CSS file: [Errno 2] No such file or directory: '/tmp/nonexistent.css'" — exit code 1.

8.5 Partial failure in multi-format

Step	Command	Expected
8.5a	<code>python -m mdutil --export bad,pdf /tmp/qa-test.md</code>	"Error: Unsupported export format: 'bad' (choose from pdf, html)" — PDF export still runs and succeeds. Exit code 1 (overall).

9. Stdin & Pipelines

9.1 Pipe Markdown to stdout viewer

Step	Command	Expected
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| 9.1a | `echo "# Hello World" \| python -m mdutil` | Rendered output: # Hello World (styled). Exits immediately. | | 9.1b | `echo "# Hello\n\nWorld" \| python -m mdutil --quiet` | No output. Exit code 0. |

9.2 Pipe Markdown to export

Step	Command	Expected
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| 9.2a | `cat /tmp/qa-test.md \| python -m mdutil --export pdf --output /tmp/pipe-test.pdf` | PDF generated. | | 9.2b | `head -c 5 /tmp/pipe-test.pdf | %PDF-` | | 9.2c | Clean up: `rm /tmp/pipe-test.pdf` | |

9.3 Multiple files

Step	Command	Expected
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9.3a	echo "Doc 1" > /tmp/doc1.md && echo "Doc 2" > /tmp/doc2.md	
9.3b	python -m mdutil /tmp/doc1.md /tmp/doc2.md	Opens first file. After quitting, opens second file.
9.3c	Clean up: rm /tmp/doc1.md /tmp/doc2.md	

10. Performance & Large Documents

10.1 Generate a large document

Step	Command	Expected
10.1a	python -c "print('# Large Doc\n'); [print(f'## Section {i}\n\nLorem ipsum dolor sit amet.\n') for i in range(200)]" > /tmp/large.md	Creates ~200-section doc (~1000+ lines).
10.1b	wc -l /tmp/large.md	~1000+ lines.

10.2 Render performance

Step	Command	Expected
10.2a	time python -m mdutil --quiet /tmp/large.md	Renders in under 2 seconds. Exit code 0.
10.2b	time python -m mdutil --export pdf --output /tmp/large.pdf /tmp/large.md	Exports PDF in under 5 seconds. File is valid.
10.2c	time python -m mdutil --export html --output /tmp/large.html /tmp/large.md	Exports HTML in under 5 seconds. File is valid.
10.2d	Clean up: rm /tmp/large.md /tmp/large.pdf /tmp/large.html	

11. Platform-Specific Notes

Linux (Ubuntu 24.04 / Fedora)

- PDF Unicode fonts (DejaVu TTF) are auto-detected from /usr/share/fonts/truetype/dejavu/
- All sections above should pass fully.

macOS

- DejaVu fonts may not be pre-installed. If missing, PDF export falls back to Helvetica/Courier (Latin-1 only). CJK/emoji characters will produce a warning (glyph missing) but the PDF will still render Latin-1 text correctly.
- To install DejaVu fonts on macOS: brew install --cask font-dejavu
- All other sections should pass.

Windows (10/11)

- Shell note: Use PowerShell, Windows Terminal, or Command Prompt. Some echo -e commands may not work — use cat and redirection instead.
- Stdin pipe: In PowerShell, Get-Content qa-test.md | python -m mdutil works.
- TTY detection: Interactive viewer requires a real TTY. PowerShell ISE may not work; use Windows Terminal.
- Configuration path: %USERPROFILE%\mdutil.ini instead of ~/.mdutilcfg.
- DejaVu fonts: May not be pre-installed. PDF export will fall back to Latin-1 fonts.
- Known limitation: file command is not available on Windows — use python -c "print(open('/tmp/test.pdf','rb').read(5))" to check for %PDF-.

12. Sign-Off

Platform	Tester	Date	Pass/Fail	Notes
Linux (Ubuntu)				
Linux (Fedora)				
macOS (Intel)				
macOS (Apple Si)				
Windows 10				
Windows 11				

Test summary: __ / 61 tests passed (__ / 61 attempted) **Overflow failures:** __ (link to bug tracker issue(s))